

A8. Create a web page containing simple calculator which should have basic arithmetic (+,-,*,/) operation on two floating point numbers and show result. Validations to be followed:

- **(Decimal point) should be taken only once for an operand.**
- **Operand can be negative.**
- **Division by zero must be shown proper error message in result.**

```
<!DOCTYPE html>
<html>
<head>
<meta charset="utf-8">
<title>Simple Calculator using HTML, CSS and JavaScript</title>
</head>
<body>
<style>
    .calculator
    {
        padding: 10px;
        border-radius: 1em;
        height: 380px;
        width: 400px;
        margin: auto;
        background-color: skyblue;
        box-shadow: rgba(0, 0, 0, 0.19) 0px 10px 20px, rgba(0, 0, 0, 0.23) 0px 6px 6px;
    }
    .display-box
    {
        font-family: 'Orbitron', sans-serif
        background-color: red;
        border: solid black 0.5px;
        color: black;
        border-radius: 5px;
        width: 100%;
        height: 65%;
    }
    #btn
    {
        background-color: red;
    }
    input[type=button]
    {
        font-family: 'Orbitron', sans-serif;
        background-color: blue;
        color: white;
        border: solid black 0.5px;
        width: 100%;
        border-radius: 5px;
```

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    height: 70%;
    outline: none;
}
input:active[type=button] {
    background: #e5e5e5; -webkit-box-shadow: inset 0px 0px 5px #c1c1c1; -moz-box-shadow:
    inset 0px 0px 5px #c1c1c1;
    box-shadow: inset 0px 0px 5px #c1c1c1;
}
</style>
<script>
    // This function clear all the values
    function clearScreen()
    {
        document.getElementById("result").value = "";
    }
    // This function display values
    function display(value)
    {
        document.getElementById("result").value += value;
    }
    // This function evaluates the expression and returns result
    function calculate()
    {
        var p = document.getElementById("result").value;
        if (p.includes('/0'))
        {
            document.getElementById("result").value = 'Division by zero Error';
        }
        else
        {
            var q = eval(p);
            document.getElementById("result").value = q;
        }
    }
    function setDot()
    {
        var result = document.getElementById("result");
        var lastval = result.value.charAt(result.value.length - 1);
        if( lastval != ".")
            result.value += ".";
    }
</script>

```

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<table class="calculator">
  <tr>
    <td colspan="3"> <input class="display-box" type="text" id="result" /> </td>
    <!-- clearScreen() function clears all the values -->
    <td> <input type="button" value="C" onclick="clearScreen()" /> </td>
  </tr>
  <tr>
    <!-- display() function displays the value of clicked button -->
    <td> <input type="button" value="1" onclick="display('1')" /> </td>
    <td> <input type="button" value="2" onclick="display('2')" /> </td>
    <td> <input type="button" value="3" onclick="display('3')" /> </td>
    <td> <input type="button" value="/" onclick="display('/')" /> </td>
  </tr>
  <tr>
    <td> <input type="button" value="4" onclick="display('4')" /> </td>
    <td> <input type="button" value="5" onclick="display('5')" /> </td>
    <td> <input type="button" value="6" onclick="display('6')" /> </td>
    <td> <input type="button" value="-" onclick="display('-')" /> </td>
  </tr>
  <tr>
    <td> <input type="button" value="7" onclick="display('7')" /> </td>
    <td> <input type="button" value="8" onclick="display('8')" /> </td>
    <td> <input type="button" value="9" onclick="display('9')" /> </td>
    <td> <input type="button" value="+" onclick="display('+)" /> </td>
  </tr>
  <tr>
    <td> <input type="button" value="." onclick="setDot()" /> </td>
    <td> <input type="button" value="0" onclick="display('0')" /> </td>
    <!-- calculate() function evaluates the mathematical expression -->
    <td> <input type="button" value="=" onclick="calculate()" /> </td>
    <td> <input type="button" value="*" onclick="display('*)" /> </td>
  </tr>
</table>
</body>
</html>

```

OUTPUT



